Usage of AI in the Project

Generative AI: A Tool Worth Using Wisely

The ICA Principle:

1. **Invite** — choose when to collaborate with AI; avoid automatic use.
2. **Control** — direct AI, review all output, and decide what to keep.
3. **Automate** — delegate repetition; own the problem-solving.

You Are the Problem Solver

AI supports effective problem-solving. You remain responsible.

- Copying without understanding eventually shows—to you, teammates, and others.
- Be the problem solver AI can't replace.
- Use AI carefully.

Why, What, How — Who Does What?

1. **Why** (worth solving?) — only you can answer.
2. **What** (core problem, minimum solution) — you decide; AI can suggest.
3. **How** (code, tests) — AI helps the most.

AI can't replace: finding solutions, communication, planning, understanding stakeholder needs, ethics, trade-offs, accountability.

AI Use Options

Options by course level:

- **200-level:** Write your own code. No vibe coding. AI may support tests, refactoring, and debugging.
- **400-level:** Vibe coding allowed; understand at least 80% of AI-generated code.
- **Any level:** AI use is optional.

What Is Vibe Coding? (200-level Examples)

Vibe coding = having AI generate the code from your prompts. Allowed at 400-level only, and you must understand what it generates.

- **OK (200-level):** ask about errors, edge cases, or bug causes; write fixes yourself.
- **Not OK (200-level):** paste an AI-written feature or module.

(When in doubt, ask the instructor.)

At 400-level, AI shifts effort from coding to ideas. Go beyond a project AI could produce in 10 minutes.

- Aim higher.
- Solve hard problems.
- Impress others.

What AI Can Help With

Use AI to support your work:

- Learning programming syntax
- Discussing architecture & design
- Making tests
- Making documentation
- Writing code (if you choose vibe coding)

Learn from AI: Programming & Syntax

Use AI for syntax, not problem-solving.

- Look up unfamiliar syntax or APIs
- Get quick explanations of error messages
- Learn idioms for a new library

Ask "why," not just "what to type."

Discuss Architecture & Design with AI

Discuss design with AI. Make decisions yourself.

- Brainstorm alternative architectures or data models
- Talk through trade-offs before committing
- Get a second opinion on your design

AI speeds up thinking; you make the call.

Making Tests

AI can generate test scaffolding.

- Draft unit tests for a module you've written
- Suggest edge cases you might have missed
- Expand test coverage for existing code

Read each test and understand what it checks.

Making Documentation

Let AI draft; edit for accuracy.

- Draft README files, API docs, or code comments
- Summarize what a module does
- Turn your notes into a clean user manual

Verify AI-written docs against the actual code.

Writing Code (If You Choose To)

If you choose vibe coding, direct AI to write code.

- Describe the feature or module you want
- Review and test what it produces before accepting it
- Understand and defend the result

Required Safeguards and Disclosure

Regardless of your course level:

- Follow this course's AI policy
- Disclose meaningful AI use — tool, purpose, and key generated artifacts
- Verify AI output with tests and review
- Never upload secrets, personal data, or restricted code

You're Still Accountable

- Explain architecture and behavior; justify key implementation decisions
- Debug and modify submitted code independently
- Own the final result, including AI mistakes

- **400-level:** meet the 80% bar; be ready to explain, debug, and modify any generated part.

AI is optional. Responsible use is evaluated when you use it.